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Diffusional Viscosity of a Polycrystalline Solid

CONYERS HERRING

Bell Telephone Laboratories, Murray Hill, New Jersey

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According to a suggestion of Nabarro, any crystal can change its shape by self-diffusion in such way as to yield to an applied shearing stress, and this can cause the macroscopic behavior of a polycrystalline solid to be like that of a viscous fluid. It is possible that this phenomenon is the predominant cause of creep at very high temperatures and very low stresses, though not under more usual conditions. The theory underlying it is developed quantitatively, and calculations of rate of creep, or equivalently of effective viscosity, are given for aggregates of quasi-spherical grains and for wires composed of cylindrical grains. Allowance is made for the effect of tangential stress relaxation at the grain boundaries. It is suggested that mosaic boundaries and boundaries between grains of nearly the same orientation may be unable to serve as sources or sinks of the diffusion currents, in which case the creep rate will depend only on the configuration of grain boundaries having a sizable orientation difference. Numerical comparison of the theory with experiments on the high temperature creep of wires favors this view, but is not entirely satisfactory. Suggestions for further experiments are made.

NABARRO¹ has recently pointed out that self-diffusion within the grains of a polycrystalline solid can cause the solid to yield to an applied shearing stress, the yielding being caused by a diffusional flow of matter within each crystal grain away from boundaries where there is a normal pressure and toward those where there is a normal tension, as shown in Fig. 1. This yielding is macroscopically describable by an effective viscosity proportional to the square of the linear dimensions of the grains. This phenomenon, which occurred independently to the present writer, provides a possible explanation for the quasi-viscous behavior which has been observed for foils and wires which are suspended with a very slight load and held at an elevated temperature.² It may also be important in other types of

sintering phenomena. The purpose of the present note is to clarify certain points in the theory underlying this "diffusional viscosity" and to attempt a more refined quantitative calculation of its magnitude for several experimentally realizable situations.

1. BASIC THEORY

Let us consider first the simplest case of stress-motivated diffusion, namely self-diffusion in a single crystal of the cubic class containing only one type of atom and held at uniform temperature. In the absence of pressure gradients the diffusive flux of atoms, due to migration of lattice vacancies or interstitial atoms, will be proportional to the gradient of the concentration of these lattice defects; in the presence of pressure gradients, however, it tends to be energetically advantageous to move the lattice defects in whichever direction will relieve the inequality of pressure. It is easily shown that the effects both of concentration gradients and of pressure gradients are correctly given by writing the diffusive flux $\mathbf{j}$ (atoms per unit area per unit time) as proportional to the gradient of a chemical potential.^{3,4} If n_L is the number of lattice sites per unit volume, D the self-diffusion coefficient, and T the absolute temperature,

$$\mathbf{j} = -(n_L D / kT) \nabla(\mu - \mu_h), \quad (1)$$

where $(\mu - \mu_h)$ is the work per atom required for the reversible addition of a small quantity of material to the given region of the crystal, the material added being placed interstitially in the lattice or used to fill lattice

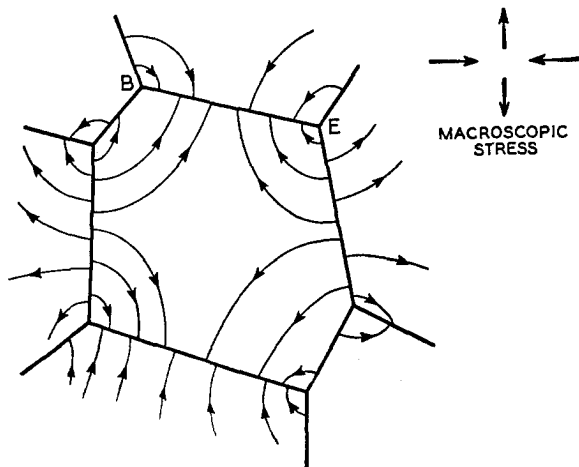


FIG. 1. A typical grain of a polycrystalline solid, showing the self-diffusion currents to be expected when the solid is subjected to a shearing stress.

¹ F. R. N. Nabarro, "Report of a Conference on the Strength of Solids" (Physical Society, London, 1948), p. 75.

² Foils: J. C. Chapman and H. L. Porter, Proc. Roy. Soc. A83, 65 (1909); I. Sawai and M. Nishida, Zeits. f. Anorg. u. Allg. Chemie. 190, 375 (1930); G. Tammann and W. Boehme, Ann. d. Physik 12, 820 (1932). Wires: Udin, Shaler, and Wulff, J. Met. 1, 186 (1949); Alexander, Kuczynski, and Dawson, paper presented at the

Symposium on the Physics of Powder Metallurgy organized by Sylvania Electric Products, Inc., New York, August 24-26, 1949.

³ It has long been recognized that it is more appropriate to write the diffusion equations for multi-component systems with chemical potentials, or their equivalents, rather than with concentrations, and this fact has recently been the subject of a number of papers in the metallurgical literature. The most recent, and probably the most satisfactory, of these is that of J. Bardeen, Phys. Rev. 76, 1403 (1949).

⁴ C. Herring, paper presented at the Symposium on the Physics of Powder Metallurgy organized by Sylvania Electric Products, Inc., New York, August 24-26, 1949. Plans have been made to publish the papers of this Symposium in book form.

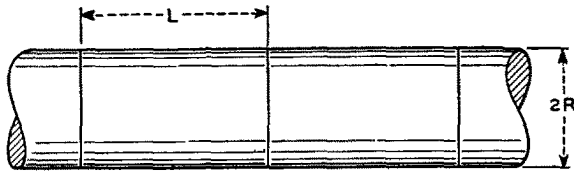


FIG. 2. Section of a wire with equally spaced transverse grain boundaries.

vacancies. This work is represented as a difference of two chemical potentials, μ and μ_h , to emphasize the fact that the change in question alters both the number of atoms and the number of lattice defects; however, in this paper only the difference of these quantities will appear.

Since any divergence of $\mathbf{j}$ within a single crystal would rapidly alter the concentration of lattice defects in such direction as to reduce $|\nabla \cdot \mathbf{j}|$, it will be safe to assume that in any continuing change $\nabla \cdot \mathbf{j} \propto \nabla^2(\mu - \mu_h) = 0$. Thus the diffusion currents are completely determined if the values of $(\mu - \mu_h)$ on the boundary of the crystal are known. Now if the transfer of atoms from the interior to lattice sites at the boundary can occur easily, i.e., if the surface of the crystal is free to grow or recede with an accompanying change in the number of lattice defects in the interior, the boundary values of $(\mu - \mu_h)$ can be determined from the condition that the free energy of the crystal and its surroundings be a minimum with respect to transfer of atoms between any portion of the surface and the volume immediately beneath this portion. In reference 4 the boundary values of $(\mu - \mu_h)$ are determined in this way for the free surface of a crystal, and for grain boundaries where the interfacial tension between the grains is important; here we shall consider the simpler case where surface free energies can be neglected in comparison with the effects of externally applied forces. In intermediate cases the two effects have merely to be added.

Assuming that the surface-volume equilibrium just mentioned is realized at a grain boundary, the work involved in adding δN atoms to a crystal grain in the form of a hump on its surface must turn out to be the same, to the first order in δN , as the work involved in putting δN atoms into the interior just below the same surface region, holding the position of the surface fixed and placing the new atoms in interstitial positions or in places formerly occupied by lattice vacancies. The latter work is $(\mu - \mu_h - \mu_0)\delta N$, where μ_0 is the chemical potential of the source from which the δN atoms are obtained, say a piece of the same substance in equilibrium at the same temperature as the crystal being considered and at zero stress. The other process, formation of a surface hump, involves an amount of work $(\delta F - \mu_0\delta N - p_{zz}\delta v)$, where δF is the free energy of the hump itself, δv its volume, and p_{zz} is the normal traction on the surface, i.e., the zz component of the stress-tensor, where the z axis is chosen normal to the surface. Now to the first order in the stresses, $\delta F = \mu_0\delta N$, so equating the

two amounts of work gives

$$\mu - \mu_h = \mu_0 - p_{zz}\Omega_0, \quad (2)$$

where $\Omega_0 = \delta v / \delta N$ is the atomic volume. This is the generalization of Nabarro's result regarding the effect of normal stress on vacancy concentration at the surface.

One can easily generalize Eqs. (1) and (2) to non-cubic crystals or to crystals containing more than one kind of atom, provided the composition is uniform. In the multi-component case, however, care must be exercised in attempting to relate the D occurring in (1) to diffusion coefficients measured in other ways. Although a detailed discussion of diffusion in multi-component systems would be beyond the scope of this paper, an attempt has been made in the Appendix to indicate the type of analysis which must be used in applying the present equations to such things as substitutional solid solutions.

2. SPECIFIC CALCULATIONS

The rate of yielding of any given specimen to applied forces obviously depends on the detailed distribution of the sizes and shapes of its crystal grains, and on whether or not the grain boundaries are able to sustain shearing stresses for times as long as are required for appreciable diffusional transport to take place. It has been established in a number of cases that shearing stresses across metallic grain boundaries are rapidly relaxed at high temperatures;^{5,6} it seems likely that this relaxation is a general property of grain boundaries, and that until this relaxation has become complete, the rate of creep due to it will usually be much faster than that due to diffusional viscosity. Although it is thus much more reasonable to compute the diffusional viscosity on the assumption of relaxed tangential stresses, we shall give calculations for both relaxed and unrelaxed tangential stresses, to show how much difference this factor makes. Both these calculations will be made assuming uniform, roughly spherical grains of a size small compared with that of the specimen; to show the way in which the rate of yielding is affected by having grains comparable with the dimensions of the specimen, some calculations will then be given for the rate of elongation of a wire consisting of roughly cylindrical grains end to end. Most of these cases are of course too idealized to apply accurately to any actual case, but it is hoped that they will sometimes be fair approximations to the truth.

Equal Quasi-Spherical Grains, No Grain Boundary Flow

If a crystal made up of grains such as that of Fig. 1 were suddenly subjected to a shearing stress, the initial diffusion currents would be very large in the neighborhoods of all sharp edges, such as E in the figure. This means that any grain boundary such as BE which is

⁵ T. S. K&, Phys. Rev. **71**, 533 (1947); J. App. Phys. **19**, 285 (1948); **20**, 274 (1949).

⁶ King, Cahn, and Chalmers, Nature **161**, 682 (1948).

subjected to a normal tension will have more extra matter squeezed in at its edges than at its center. This will increase the tension at the center and decrease that at the edges, and a situation will very soon be reached where the normal flux is fairly uniform over each facet of the boundary. This suggests that for a crystal made up of nearly spherical grains all of the same size and having nearly isotropic elastic properties the rate of yielding can be fairly well approximated by that of a sphere acted on by such surface tractions as will produce a uniform shearing stress in its interior.

We shall assume for simplicity that the equations of Section 1 for a cubic crystal apply. Let the macroscopic stress-tensor be $\bar{p}_{\mu\nu}$. This must in all cases equal the volume average of the microscopic stress; in the present case we assume the stress in the interior of the sphere to be constant and equal to $\bar{p}_{\mu\nu}$. We assume $\bar{p}_{\mu\nu}$ to be a pure shear, so that

$$\sum_{\mu} \bar{p}_{\mu\mu} = 0.$$

At a point of the surface of the sphere with coordinates x_{μ} the normal traction is

$$p_{rr} = \sum_{\mu, \nu} \bar{p}_{\mu\nu} x_{\mu} x_{\nu} / R^2,$$

where R is the radius of the spherical grain, so that

$$\sum_{\mu} x_{\mu}^2 = R^2.$$

Inserting this into the boundary condition (2) it is easily seen that since $\nabla^2(\mu - \mu_h) = 0$ the chemical potential in the interior must be given by

$$\mu - \mu_h = \mu_0 - \Omega_0 \sum_{\mu, \nu} \bar{p}_{\mu\nu} x_{\mu} x_{\nu} / R^2. \quad (3)$$

The normal flux at the boundary is, by (1)

$$j_n = (dr/dt) / \Omega_0 = (2D/kT) \sum_{\mu, \nu} \bar{p}_{\mu\nu} x_{\mu} x_{\nu} / R^3 \quad (4)$$

since $n_L \Omega_0 = 1$. This may be compared with the rate of normal displacement of the boundary in a uniform shear of strain rate $\dot{\epsilon}_{\mu\nu}$:

$$dr/dt = \sum_{\mu, \nu} \dot{\epsilon}_{\mu\nu} x_{\mu} x_{\nu} / R. \quad (5)$$

Comparison of (4) and (5) shows that our assumptions are consistent with a uniform shear of all the grains of the crystal at a rate

$$\dot{\epsilon}_{\mu\nu} = (2D\Omega_0/kTR^2) \bar{p}_{\mu\nu}. \quad (6)$$

Viscous flow is defined, for the present case where $\bar{p}_{\mu\nu}$ is a pure shear, by the macroscopic relation $2\dot{\epsilon}_{\mu\nu} = \bar{p}_{\mu\nu}/\eta$, which with (6) gives for the effective viscosity of the polycrystalline solid

$$\eta = kTR^2/4D\Omega_0 = (3/32\pi)^{1/2} (kT/D\Omega_0) V_g^{1/2}, \quad (7)$$

where V_g is the volume of a grain.

Equal Quasi-Spherical Grains, Tangential Stress Relaxed at Boundaries

For this case we must find a distribution of normal tractions over the surface of a sphere which will make the volume average of the stress-tensor in the interior equal the macroscopic stress-tensor $\bar{p}_{\mu\nu}$. Now from the equilibrium condition that the divergence of the stress-tensor must vanish we have, if $d\tau$ runs over the volume of the sphere and dS over its surface

$$\begin{aligned} 0 &= \int x_{\mu} \sum_{\alpha} \partial p_{\alpha\nu} / \partial x_{\alpha} d\tau \\ &= - \int p_{\mu\nu} d\tau + \int x_{\mu} \sum_{\alpha} p_{\alpha\nu} n_{\alpha} dS, \end{aligned} \quad (8)$$

where $\mathbf{n}$ is the unit outward normal. For the present case

$$\sum_{\alpha} p_{\alpha\nu} n_{\alpha} = p_{nn} n_{\nu},$$

so this gives

$$\bar{p}_{\mu\nu} = (3/4\pi R^4) \int x_{\mu} x_{\nu} p_{nn} dS. \quad (9)$$

If p_{nn} is represented by a series of spherical harmonics, (9) will determine the coefficients of the harmonics of order 2 (the lowest order consistent with the conditions of the problem) but will leave the higher order terms arbitrary. In an actual crystal grain these higher order terms will probably be no larger than is required to take account of the departure of the grain boundary configuration from spherical symmetry, and it is therefore proper to neglect them in the present approximation, the more so as they make no contribution to the over-all shear rate. Let us therefore set $p_{nn} = A(x_1^2 - x_2^2)/R^2$, a form which can always be realized for the contribution of the second order spherical harmonics by proper choice of the coordinate directions x_1 and x_2 . Insertion of this into (9) gives $\bar{p}_{11} = -\bar{p}_{22} = 2A/5$, all other $\bar{p}_{\mu\nu} = 0$. The chemical potential in the interior of the sphere is given by

$$\mu - \mu_h = \mu_0 - (5/2) \bar{p}_{xx} \Omega_0 (x^2 - y^2) / R^2. \quad (10)$$

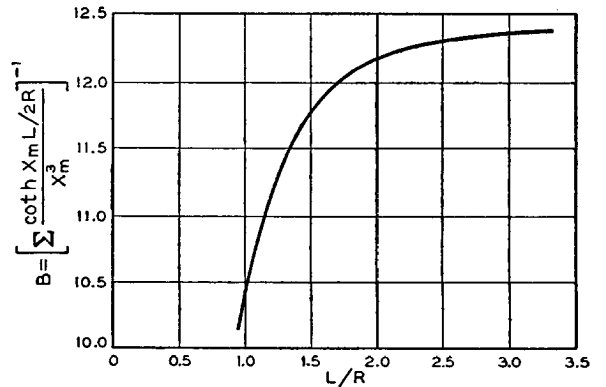


FIG. 3. Plot of the coefficient B occurring in Eq. (18) against the ratio of the length L of the grains to the radius R of the wire.

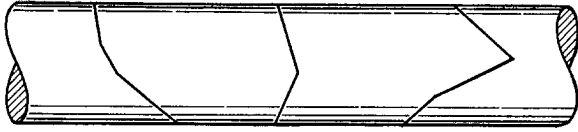


FIG. 4. Wire with irregular grain boundaries.

This is of the same form as (3) except for the factor $5/2$. Repeating the argument leading to (7) gives

$$\eta = kTR^2 / \frac{1}{10} D\Omega_0 = \frac{1}{10} (3/4\pi)^{1/2} (kT/D\Omega_0) V_0^{1/2}, \quad (11)$$

which is smaller than (7) by a factor $\frac{2}{5}$.

Wire with "Bamboo-Like" Grain Structure

Fine metal wires often consist of crystal grains occupying the entire cross section of the wire and having lengths which may be of the same order as the diameter of the wire or may be many times larger. In the experiments cited in reference 2 this type of grain structure characterized both the copper wires used by Udin, Shaler, and Wulff and the gold wires used by Alexander, Kuczynski, and Dawson.* A model which approximates the condition of these wires and for which the associated potential theory problem is easily solved is that of a wire with a "bamboo-like" grain structure, i.e., a cylinder of radius R with plane grain boundaries normal to the axis of the cylinder, uniformly spaced a distance L apart, as shown in Fig. 2.

For this model it is obvious from symmetry that when a longitudinal tension or compression is applied, the

diffusion currents in the quasi-steady state will be such as to make the normal flux away from or into each grain boundary a constant over its area. This will of course require that the normal traction p_{nn} across a grain boundary vary with distance from the center of the wire; however, the integral condition

$$\int p_{nn} dS = F$$

must always be obeyed, where dS runs over the area of a grain boundary, and F is the force applied at the ends of the wire. For simplicity we ignore surface tension effects for the present; this is legitimate in computing the change in rate of stretching with a change in the load. In such case the boundary conditions on $(\mu - \mu_h)$ are:

$$f \equiv \mu - \mu_h - \mu_0 = 0 \quad \text{on the cylindrical surface} \quad (12)$$

while over the end boundary of each grain

$$\partial f / \partial z = \text{constant} \quad (13)$$

$$\int f dS = -F\Omega_0 \quad (14)$$

where z is the coordinate in the direction of the axis of the cylinder. The determination of f involves the same problem, therefore, as the determination of the potential due to an infinite series of uniformly charged disks placed inside a conducting cylinder. Using standard methods⁷ the solution is easily shown to be

$$f = - \frac{F\Omega_0 \sum_m [\coth(X_m L/2R) \cosh(X_m z/R) - \sinh(X_m z/R)] J_0(X_m r/R) / X_m^2 J_1(X_m)}{2\pi R^2 \sum_m \coth(X_m L/2R) / X_m^3} \quad (15)$$

where the J 's are Bessel functions, X_m runs over the roots of $J_0(X_m) = 0$, and r is the distance from the axis of the cylinder, and where z ranges from 0 on one of the grain boundaries to $L/2$ midway between this boundary and the next. Each grain elongates or contracts at a rate

$$dL/dt = - (2\Omega_0 / \pi R^2) \int j_z dS,$$

the integration being over the disk of radius R in the plane $z=0$. Using (1) and (15) we find

$$dL/dt = (D\Omega_0 / 2\pi R^3 kT) BF, \quad (16)$$

where

$$B = \left[\sum_m \coth(X_m L/2R) / X_m^3 \right]^{-1} \quad (17)$$

is a function of L/R plotted in Fig. 3. Its asymptotic value for $L \gg R$ is $B_\infty = 12.37$, and it does not begin to depart seriously from this value until L/R has become

* I am indebted to the authors of these two papers for discussions of unpublished details of their work, including opportunities to examine photomicrographs of their wires, and for the statistics on mean grain lengths which have been used in Table I.

smaller than any value likely to be encountered frequently in wires whose crystal grains occupy the whole cross section.

In the present case the deformation of the wire cannot legitimately be described as viscous, because the grain size equals or exceeds the diameter of the specimen; however, it is interesting for comparative purposes to compute the viscosity η which a viscous filament of the same diameter would have to have in order to elongate at the same rate under the same load. For such a filament we have $F = \pi R^2 \cdot 3\eta d \ln L / dt$, and this agrees with (16) if

$$\eta = 2kTRL / 3BD\Omega_0. \quad (18)$$

Wire with Irregular or Slanting Grain Boundaries, Tangential Stresses Relaxed

It is easy to show that if the grain boundaries are not normal to the axis of the wire as in Fig. 2, but are of the type shown in Fig. 4, the rate of yielding of the wire will

⁷ See for example, W. R. Smythe, *Static and Dynamic Electricity* (McGraw-Hill Book Company, Inc., New York, 1939), Chapter 5.

be greater than that given by (16) for the same grain size, if the tangential stresses vanish at the grain boundaries. This is most easily seen by noting that a given dL/dt requires a given total normal flux integrated over the grain boundary,[†] regardless of its shape or inclination; in the electrostatic analogue this means a given total charge. The potential produced by this given charge, averaged over the grain boundary, will be numerically smaller when the grain boundary is oblique than when it runs normally across the wire as in Fig. 2. This potential is proportional to the normal stress across the grain boundary. It is easy to see that when the normal tractions are resolved along the axis of the wire, equal mean normal stresses will give equal values of the force F tending to elongate or compress the wire; therefore F will be less for Fig. 4 than for Fig. 2, when dL/dt is the same.

The calculations of the preceding section suggest some approximations which can be used to make a crude quantitative estimate of this effect. The same methods which were used to obtain (15) can be applied to calculate the potential due to a uniformly charged flat ring of any inner radius, r_1 , and outer radius r_2 , placed inside a grounded conducting cylinder. The potential problem for a grain boundary consisting of a series of ring-like steps, as shown in Fig. 5, can therefore be solved, and if the steps are not taken too large, this solution can probably be regarded as typical of irregular grain boundaries of the same average inclination. Ignoring interactions between the two ends of a grain, which Fig. 3 shows to be quite small, the force F required to produce a given elongation dL/dt at a single grain boundary of the sort described comes out to be

$$F = \frac{2\pi R^3 kT}{D\Omega_0} \times \left[\sum_m \frac{\sum_{i,j} \{m,i\} \{m,j\} \exp(-|z_i - z_j| X_m/R)}{X_m^5 [J_1(X_m)]^2} \right] \frac{dL}{dt}, \quad (19)$$

where i and j label the various rings, the i th ring having inner radius r_{i1} and outer radius r_{i2} and longitudinal coordinate z_i , and where

$$\{m,i\} = (X_m r_{i2}/R) J_1(X_m r_{i2}/R) - (X_m r_{i1}/R) J_1(X_m r_{i1}/R).$$

Now for large L the series in (17) differs from its leading term by only about 10 percent, so it is probably legitimate, in an approximation which replaces the grain boundaries of Fig. 4 by those of Fig. 5, to set the ratio of (19) to the F_0 given by (16) equal to the ratio of the first terms of the respective summations on m . This gives

$$F/F_0 \approx \sum_{i,j} \{1,i\} \{1,j\} / X_1^2 [J_1(X_1)]^2. \quad (20)$$

[†] In accordance with the discussion given in Section 4, it is assumed that for sufficiently small stresses no "offsetting" of the grains will occur.

Some values computed from this formula for typical grain boundaries are given in Fig. 5.

The conclusion from this analysis is that the equivalent viscosity of a wire with irregular grain boundaries may be as much as two to three times smaller than the value given by (18) for flat-ended grains of the same length.

3. GRAIN BOUNDARIES VERSUS MOSAIC BOUNDARIES

The calculations of the preceding section have assumed the existence of certain "grains," at whose boundaries diffusion currents of lattice defects can converge or diverge, the boundary concentration of lattice defects being in equilibrium with the normal stress. It is conceivable that these properties might be possessed by crystal grains in the usual sense of the term, by mosaic blocks within a "single crystal," or, perhaps, by neither. In the present section it will be suggested as particularly likely that the condition of equilibrium just mentioned is attained, at sintering temperatures, on most ordinary grain boundaries, but is not attained on the boundaries of mosaic blocks. The experimental evidence discussed in the next section, though not entirely satisfactory, tends to favor this suggestion.

The boundaries of mosaic blocks have been pictured as consisting of rows of identical parallel dislocations spaced apart by a distance of the order of 10^8 atomic spacings.⁸ Although it is not known whether the usual mosaic structure of single crystals consists of blocks bounded by such rows of dislocations, or merely of an irregular tangle of dislocation lines, it can hardly be doubted that the structure can be described in terms of dislocations; moreover, there seem to be good prospects for accounting quantitatively for many of the properties of grain boundaries, at least when the orientation difference between the grains is small, by picturing the boundaries as arrays of dislocations.⁹ It is therefore

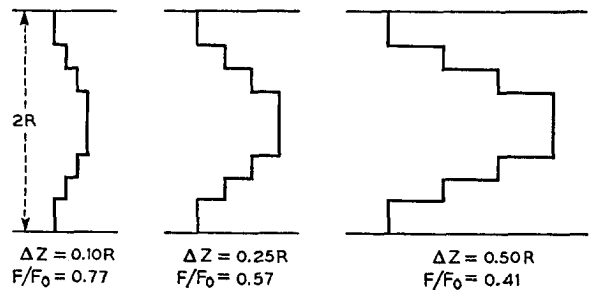


FIG. 5. Representation of irregular grain boundaries by successions of ring-like steps of height Δz at radii $0.3R$, $0.5R$, and $0.7R$. The quantity F/F_0 shown for each case is the approximate ratio of the force required to produce a given rate of elongation of the grain shown to the force required to produce the same rate of elongation of one of the flat-ended grains of Fig. 2.

⁸ J. M. Burgers, Proc. Phys. Soc. **52**, 23 (1940); J. S. Koehler and F. Seitz, J. App. Mech. **14**, A217 (1947).

⁹ W. Shockley and W. T. Read, Phys. Rev. **75**, 692 (1949); W. T. Read and W. Shockley, Phys. Rev. **78**, 275 (1950).

legitimate to wonder what difference, if any, there is between the behavior of grain boundaries and that of mosaic boundaries.

In a well-annealed crystal the configurations of the dislocation arrays describing the mosaic structure must be such as to make the elastic free energy a minimum with respect to small displacements of any of the dislocation lines. Application of a small stress will affect the dislocations in two ways: a shearing stress on the slip plane of a dislocation will tend to move it in this plane, while a normal stress across a plane perpendicular to the slip vector will tend to move the dislocation in a direction not lying in the slip plane. Motion of the latter type requires the creation or annihilation of lattice defects, and will cease when the chemical potential of the lattice defects in the neighborhood of the dislocation has changed by an amount sufficient to compensate for the change in stress. An isolated dislocation line would thus affect the potential ($\mu - \mu_h$) of the diffusion problem in much the same way as a wire held at a certain potential would affect an electrostatic problem. However, in an array of reasonably well-separated dislocations it seems likely that the freedom of motion of each dislocation will be impeded by the stress-fields of neighboring dislocations, and that within a short time after application of a small stress a new elastic equilibrium will be reached and all motion of the dislocations will cease, and concomitantly ($\mu - \mu_h$) will become essentially constant over mosaic dimensions.

One aspect of this speculation can be given a rough check. The distance which each dislocation will move, in coming to a new position of equilibrium after application of a small stress σ , may be expected to be of the order of magnitude of $\sigma d / \sigma_0(d)$, where d is a length of the order of the spacing between dislocations and $\sigma_0(d)$ is the stress at a distance d from an isolated dislocation line. The resultant shearing strain of the whole specimen should therefore be of the order of this quantity times a/d^2 , where a is the lattice constant. Since $\sigma_0(d)$ is of the order of $Ga/2\pi(1-p)d$, where G is a shear constant and p a Poisson's ratio,¹⁰ the resulting shear is of order $2\pi(1-p)\sigma/G$. This gives values two or three orders of magnitude smaller than the total strains observed in the wire experiments of reference 2, which were of the order of 10^{-2} .

It is therefore not implausible to suppose that motion of dislocations may be of negligible importance in these experiments, and that the diffusion currents are divergenceless except at grain boundaries separating two crystals having a sizable difference of orientation $\Delta\theta$. By "sizable" is meant an orientation difference large enough so that the grain boundary can no longer be viewed as an array of dislocations spaced many lattice constants apart. If this speculation should turn out to be true, it would imply the existence of a critical value of $\Delta\theta$ below which growth of the two crystals normal to the grain

boundary cannot take place freely, but above which it can. In such case any grain boundaries separating grains with less than the minimum $\Delta\theta$ should not be counted as grain boundaries in comparing the present theory with observations. There may conceivably be many such boundaries in wires whose grains have a strong tendency to crystallize with a certain crystallographic direction along the axis of the wire. For two grains with such a common axis need only a relative rotation about the wire axis to bring their orientations into coincidence, and it is possible that at sintering temperatures the natural desire of the grain boundary to decrease its interfacial tension may bring about such a rotation, at least until the orientation difference becomes small enough so that further changes of configuration cannot take place freely.

4. RELATION OF THEORY TO EXPERIMENT

Most of the present section will be devoted to a discussion of experiments on the longitudinal creep of wires at high temperatures, as measured by Udin, Shaler, and Wulff² for copper and by Alexander, Kuczynski, and Dawson² for gold. In these experiments an attempt was made to find what value of the applied tension would just suffice to prevent the wires from shrinking under the action of their surface tension, and thus to measure the latter quantity. The stresses used in these experiments were therefore below the range used in most ordinary creep experiments (about 1 to 100 g/mm²), while the extensions were relatively large, of the order of 10^{-3} to 10^{-2} and more. It is therefore not unreasonable to attribute the creep in these high temperature experiments to mechanisms quite different from those operative in most of the previous creep experiments in the literature, although it is of course quite conceivable that the mechanisms may be the same, the thresholds being lower and the rates faster at the higher temperatures. The latter possibility is supported by the fact that Miller¹¹ has observed an apparently unlimited creep for single crystals of zinc under stresses of only 2 or 3 g/mm², even at room temperature; however, it would be risky to conclude from this that the experiments of reference 2 involve the same mechanisms as room-temperature creep in single crystals, and the following discussion will explore the alternative that the diffusion mechanism discussed in the previous sections plays a major role. As Nabarro has shown,¹ the latter mechanism does not seem to be the major one in creep experiments such as those of Chalmers¹² on single crystals of tin.

Offsetting and Kinking

Besides the diffusion effect and the possible effect of ordinary creep within the individual crystal grains, there are at least two other effects which may influence the changes in length which are observed for fine wires

¹⁰ See for example, J. S. Koehler, Phys. Rev. **60**, 397 (1941).

¹¹ R. F. Miller, Trans. A.I.M.E. **122**, 176 (1936).

¹² B. Chalmers, Proc. Roy. Soc. **A156**, 427 (1936).

suspended at high temperatures. One of these is "offsetting" of adjacent grains, as shown in Fig. 6. This is known to occur for polycrystalline wires suspended at high temperatures under tension of sufficient magnitude.¹³ However, it is easy to see that it cannot occur if the tension is below a certain limit, and that it cannot contribute to the shrinking of wires suspended with no load. For, referring to Fig. 6, consider a lateral offset of amount AB along a grain boundary whose normal makes an angle θ with the axis of the wire. This enables the wire to yield to the applied tensile force F by a distance $\langle AB \rangle \sin \theta$; however, it exposes an area of new surface represented by AB and CD . The surface free energy thus increases by $2R \cdot \langle AB \rangle \cdot (2\bar{\gamma} - \gamma_b)$ where $\bar{\gamma}$ is an average surface tension for the regions AB and CD , and γ_b is the specific free energy of the grain boundary. Thus the over-all free energy will be lowered by the offset if and only if

$$F > 2R(2\bar{\gamma} - \gamma_b) \csc \theta. \quad (21)$$

This is somewhat larger than the force $\pi R \bar{\gamma}'$ necessary to prevent shortening of a long grain under its own surface tension. We may therefore conclude that offsetting will not occur in the lower range of stresses employed in the wire experiments of reference 2, but that it may well occur in the higher part of the stress range.

The other possible perturbing effect, which will be called "kinking," consists in the development of a slight angle between the axes of adjacent grains of the wire. This could be produced by a relative rotation about an axis normal to the wire axis. Such a rotation might be motivated by a dependence of the free energy of the grain boundary on the relative orientations of the grains; it would of course require that material be removed from the portions of the grains which move toward each other, and added to the portions which move apart. If this transport of matter takes place by diffusion, one can estimate the order of magnitude of the rate of change of orientation by the methods described in reference 4. This estimate, the details of which will not be given here, shows that for the wires of reference 2 the rate of kinking due to this cause ought to be too small to produce a perceptible shortening, though possibly not too small to give a measurable angle of kink.

Kinking might conceivably also be caused by a relative rotation of two grains about the normal to a plane grain boundary, if this normal were a trifle inclined to the axis of the wire. The motivation could again come from an anisotropy of the grain boundary energy; for this process, however, no transport of matter would be required, and its rate might be faster than that of the process described in the preceding paragraph.

Calculated versus Observed Creep Rates

Table I gives a comparison of the observed rates of creep of the wires of reference 2, expressed in terms of

¹³ See for example, C. J. Smithells, *Tungsten* (D. Van Nostrand Company, Inc., New York, 1936), second edition, p. 81.

equivalent viscosities, with rates calculated from (18) and (20). Since the least accurately known of the experimental data are the lengths L of the crystal grains, the comparisons are made on this quantity.

Ideally, if the diffusion hypothesis is the correct one, it should be possible to get a fairly accurate check between theory and experiment. For, with one possible exception, all the quantities on the right of (18) can be measured fairly accurately, the most uncertain being D , and even this can be measured to an accuracy of 20 percent or better if the temperature can be accurately reproduced. The possible exception is the mean length L of the grains: if, following the possibility suggested in Section 3, it turns out that growth can occur at grain boundaries where the orientation difference is large but not at those where it is small, then the value of L to be inserted into (18) will be larger than the value L' measured metallographically, by an amount which it may be hard to establish. The correction factor F/F_0 will in many cases be close to unity, since both theory and observation agree that grain boundaries usually prefer to align themselves normal to the wire axis; if for any reason the boundaries are not transverse, a reasonable estimate of F/F_0 can be made from Fig. 5.

A number of experimental uncertainties in the present examples make the accuracy of the comparison of theory and observation fall somewhat short of this ideal. Besides the uncertain possibility that L may be appreciably greater than L' , the value of L' itself is somewhat uncertain. The values given in the table were taken from photomicrographs of short sections of the wires after etching; however, since regions of long grains may alternate with regions of short grains, many such samples must be measured to get a reliable measure of the mean length of all the grains in the specimen. Moreover, the grain structure may change appreciably during the experiment. In this connection it is interesting to speculate that a relative axial rotation of neighboring grains, resulting ultimately in making the orientation difference too small for growth to occur, might conceivably cause the effective L to increase with time. This would lead to a creep rate decreasing with time, as was observed by Alexander, Kuczynski, and Dawson.²

A moderate error in the temperature, or rather, in the relative temperature scales of the creep and diffusion measurements, can produce an appreciable error in D . Dr. Alexander has informed the author that the temperature listed in the table as 1050° may have been closer to 1025° ; a temperature lowering of this amount would

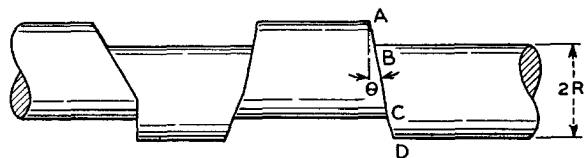


FIG. 6. Illustration of offsetting for a wire suspended under tension.

TABLE I. Comparison of observed viscosities of wires with predictions of Eqs. (18) and (20).*

Authors†	USW		AKD
	Cu	Cu	Au
Metal			
$T^{\circ}\text{C}$	1000	1000	1050
R , microns	36	64	14
D , $\text{cm}^2/\text{sec}.$ ‡	2.5×10^{-9}	2.5×10^{-9}	8×10^{-9}
Observed η , poises	4.4×10^{12}	3.2×10^{12}	1.4×10^{12} (initial) 2.4×10^{12} (final)
$(L/R)(F/F_0)$, calculated	1.0	0.24	9.6 to 16
L'/R , observed	4	2	2

* The observed η -values for copper were taken from lines drawn by Udin, Shaler, and Wulff averaging their measurements at different temperatures. Their values must be multiplied by 2 because of an error in a formula, derived by Frenkel, by which they reduced their observations.

L is the average spacing between grain boundaries at which growth can occur, while L' is the average spacing between observable boundaries.

F/F_0 is the correction factor to be applied to (18) due to departure of the grain boundary shapes from the ideal transverse planes of Fig. 2; according to (20) this factor is probably between $\frac{1}{2}$ and 1.

The next to the last line of the table was computed by equating the observed viscosity η to the product of F/F_0 by the right of (18) taking B to be 12.2, and solving for $(L/R)(F/F_0)$.

† See reference 2.
‡ The self-diffusion coefficient tabulated for copper is close to the values measured by Steigman, Shockley, and Nix, *Phys. Rev.* **56**, 13 (1939); M. S. Maier and H. R. Nelson, *Trans. A.I.M.E.* **147**, 39 (1942); and Raynor, Thomassen, and Rouse, *Trans. Am. Soc. Metals* **30**, 313 (1942). That for gold was obtained by extrapolating the data of H. A. C. McKay, *Trans. Faraday Soc.* **34**, 845 (1938).

lower D , and therefore the calculated $(L/R)(F/F_0)$, by a factor of about 1.5.

If the diffusion of a radioactive tracer isotope occurs largely or predominantly by simultaneous interchange of positions by neighboring atoms, the D to be used in the present calculations will of course be less than that measured in the tracer experiment, perhaps very much less.

Remarks on Future Experiments

The comparisons given in Table I, though not very satisfactory, do at least show that the creep observed in the experiments of reference 2 is of the same order of magnitude as that predicted on the hypothesis that it is due to diffusion currents flowing between the free surface and the grain boundaries and that it is therefore distinctly slower than that which would be predicted by an analogous theory in which the diffusion currents converge on the boundaries of mosaic blocks with dimensions of the order of 10^{-4} cm. However, an obvious direct check of the hypothesis that grain boundaries are essential could be made by performing the same experiments with single crystal wires, or with wires having a very large L/R ratio. In such an experiment it would be desirable to check the orientations of different parts of the "single crystals" by x-ray methods, to detect the possible presence of a polycrystalline structure with grain orientations differing by only a few degrees.¹⁴

Several items may be mentioned which would facilitate the interpretation of future experiments with the usual polycrystalline wires. Grain lengths should be measured over the whole length of the wire, and if possible the distribution of orientation differences be-

¹⁴ Such structures have been observed in a "single crystal" of aluminum by P. Lacombe, "Report of a Conference on Strength of Solids" (The Physical Society, London, 1948), p. 91.

tween adjacent grains should be checked. Checks should also be made to find out whether the sizes and orientations of the grains change appreciably during the experiment, and whether changes in the creep rate with time are permanent or recoverable. Offsetting and kinking should of course be watched for, though according to the discussion above they are not expected to play a major role in the stress range of most interest.

Diffusional viscosity should not be very greatly affected by small amounts of impurities, once the effect of these impurities on the grain structure has been allowed for. This could easily be checked.

An interesting consequence of the theory of this paper is that the cylindrical surfaces of the wires used in experiments such as those of reference 2 ought to develop shallow depressions along the grain boundaries when they lengthen with time, and ridges when they shrink. This is because the diffusion currents tend to be especially strong at the parts of the surface near the grain boundaries. Use of polished wires would facilitate detection of this effect.

In conclusion, it should be pointed out that the phenomenon of diffusional viscosity, if it turns out to be the predominant cause of creep under very low stresses at sintering temperatures, should be very important in all kinds of sintering processes. This is especially true with regard to the final stages of sintering: the shrinking down of the cavities requires an overall contraction of the specimen, and this can easily be achieved if the crystal grains are free to slide on each other tangentially and to grow or recede normally to their boundaries. In the absence of such processes and of plastic flow within the crystals, the only mechanism which could shrink the cavities would be a diffusion of material from the outer boundary of the specimen, as postulated in the theory of Pines.¹⁵ The latter process obviously gives a rate of densification much slower than the former process, and one which decreases rapidly with increasing dimensions of the specimen.

APPENDIX

In a cubic crystal containing various kinds of atoms, the general expression for the diffusive flux of the i th kind of atom is

$$\mathbf{j}_i = -(n_L/kT) \sum_k D_{ik} \nabla(\mu_k - \mu_h), \quad (\text{A})$$

where n_L is the number of unit cells in unit volume. Immediately after stresses are applied, the currents of the different species will in general be widely different; however, before any macroscopically recognizable transport of material has occurred concentration gradients will be set up in such directions as to tend to equalize these differences in currents, and if the stresses are small a quasi-steady state will very soon be reached in which $\mathbf{j}_i = X_i \mathbf{j}$, where X_i is the fractional concentration of the i th species and $\mathbf{j} = \sum \mathbf{j}_i$. It will be shown that $\mathbf{j}$ can be expressed in terms of the gradient of a single harmonic function. Equating (A) to $X_i \mathbf{j}$ and solving for $\nabla(\mu_k - \mu_h)$,

$$\nabla(\mu_k - \mu_h) = -(kT/n_L) \mathbf{j} \sum_i (D^{-1})_{ki} X_i, \quad (\text{B})$$

¹⁵ B. Y. Pines, *J. Tech. Phys., U.S.S.R.* **16**, 737 (1946). I am indebted to Dr. G. C. Kuczynski for a translation of this paper.

where D^{-1} is the reciprocal of the matrix D . Now define μ_{0k} at each point of the crystal as the value of μ_k corresponding to the composition occurring at that point but at zero stress. Then by the Gibbs-Duhem relation, $\sum X_k \nabla \mu_{0k} = 0$. Moreover, if the stresses are small ($\mu_k - \mu_h - \mu_{0k}$) will be a small quantity of the first order in the stresses. Multiplying (B) by X_k and summing gives

$$-(kT/n_L) \sum_{i,k} (D^{-1})_{ik} X_i X_k = \sum_k X_k \nabla (\mu_k - \mu_h - \mu_{0k}) \approx \nabla \sum_k X_k (\mu_k - \mu_h - \mu_{0k}) \quad (C)$$

if we neglect quantities of the second order in the stresses, since ∇X_k is a small quantity of the first order. Thus we can write, in formal analogy to (1),

$$\mathbf{j} = -(n_L D' / kT) \nabla f \quad (D)$$

with

$$f = \sum_k X_k (\mu_k - \mu_h - \mu_{0k}) \quad (E)$$

and

$$D' = [\sum_{i,k} (D^{-1})_{ik} X_i X_k]^{-1}. \quad (F)$$

An argument of the same type as that leading to (2) shows that at the boundary of the crystal

$$f = -p_{ss} \sum_k X_k (\partial v / \partial N_k) = -p_{ss} \Omega_0, \quad (G)$$

where $\partial v / \partial N_k$ is the partial atomic volume of the k th component and Ω_0 is the mean atomic volume.

The form of the matrix D_{ik} of course depends on the mechanism of diffusion, a subject beyond the scope of this paper. We may merely mention that for purely interstitial diffusion the matrix is diagonal, so that for a two-component crystal

$$D' = (X_1^2 / D_{11} + X_2^2 / D_{22})^{-1},$$

while for a two-component crystal in which there is no diffusion except by place interchange the determinant $|D_{ik}|$ vanishes, making $D' = 0$, as is to be expected for this model.

Streak Photography

IRWIN VIGNESS AND R. C. NOWAK

Mechanics Division, Shock and Vibration Branch, Naval Research Laboratory, Washington, D. C.

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A description is given of some methods of "streak photography" used for obtaining displacement records as functions of time. Ordinary streak photography consists of photographing a moving object on a moving film camera with an open shutter. The motion of the object should be in a plane parallel to the film and perpendicular to its motion. Large errors in velocity determinations may occur if the motion of the object has components parallel to the film motion. A modified method of streak photography is described in which these errors are reduced to small values. By this method time-displacement curves have been recorded for displacements of many inches and periods of several seconds with accuracies within a few thousandths of an inch and a few hundred thousandths of a second.

INTRODUCTION

THE measurement of transient motions of mechanical elements is of constant necessity in laboratory and field problems associated with mechanical shock and vibration. The conditions of the experiments require methods of measurements that are easy to perform and that do not appreciably load the mechanical parts under study. While high speed moving pictures provide this together with a wealth of other information, the work of transcribing the motion picture records into quantitative displacement curves is too laborious for many conditions. Methods of streak photography are easy to perform and can often be done with ordinary laboratory equipment. Many types of oscillographic recording cameras can be used with little or no modification. Several Naval Laboratories have made extensive use of the methods of streak photography to conveniently obtain accurate displacement curves.¹ However, no descriptions have been made of these methods except as incidental to other experiments. This paper describes the methods of streak photography most generally used.

¹ Vigness, Kammer, and Holt, "Shock and vibration instrumentation and measurements," Second Partial Report, Naval Research Laboratory Report No. 0-2645 (September, 1945).

ORDINARY STREAK PHOTOGRAPHY

One of the simplest methods of recording transient motions of an object is to photograph the object with a moving film camera. The film of the camera is made to move at a known speed and the shutter is held open during the entire exposure. The camera is arranged so that the direction of motion of the object under study is perpendicular to the camera optic axis and to the direction of film motion. Sufficiently good contrast can often be obtained without any special preparation of the object, but usually it is advisable to arrange bright spots with a dark background on the object whose motion is being studied. These bright spots may be dots of white paint, ball bearings, or small luminous sources. Ball bearings are particularly advantageous if very fine lines are desired. A small light source some distance from the ball should be used, the reflection of this light by the ball closely approximates a point. The ball should be small so that the reflected light will continue to come from essentially the same location on the ball. Many spots may be in the field of view, and, consistent with the requirements of depth of focus, they can be spread over any distance from the camera. Of course the displacement scale will depend upon this distance. Illumination